

	Yıldız Technical University- Mathematics Department Make Up Exam - Questions Sheet				Grade Table			
					1.Q		2.Q	
Name, Surname								
Student ID No			Group No	1				
Section		Mathematics				Exam Date		22.05.2020
Name of Lecture		MAT2042MATHEMATICAL ANALYSIS IV			Duration	90 mnts		
Lecturers		Asst. Prof. Elif DEMİR						
Students who cheat or even attempt to cheat in the exam are going to be punished with suspension for one or two semesters according to the Article 9 of the Student Discipline Ordinance (No.2547) of Higher Education Council (YÖK).								

1) **R:** $x = 1$, $x = 2$, $y = x$, $xy = 1$ is given.

i) **Evaluate** the integral $I = \iint_R \frac{x^2}{y^2} dydx$ on the region R.

ii) Rewrite the integral I as the order of integration reversed. (**Do not Evaluate !**)

iii) Rewrite the integral I by using polar transformation. (**Do not Evaluate !**)

2) **V :** $x^2 + y^2 + z^2 = a^2$, $x^2 + y^2 + z^2 = b^2$ is given for $a < b$.

Evaluate $I = \iiint_V \frac{1}{(x^2 + y^2 + z^2)^{3/2}} dv$

by using spherical coordinates.

3) Search that the integral $I = \int_0^\infty \frac{dx}{\sqrt{x}e^{x^2}}$ is convergent or divergent. If it is convergent, evaluate it by using gamma-beta functions.

Note: The exam will start at 10:00 and be over at 11:30. You have to send your answer sheets by a pdf or jpeg

file by attaching to an e-mail to edemirbilek@yahoo.com until 11:45 (I will not receive your answers later than this

time). Subject of your mail must be in form as “Analiz4_Name_Surname” for example Analiz4_Elif_Demir. Your

name, surname, school number and signature must be on your paper (otherwise I will not receive your answers

Good Luck..

Not: Sınav saat 10:00 da başlayacak ve 11:30 da bitecektir. Cevap kağıtlarınızı pdf veya jpeg dosyası olarak

edemirbilek@yahoo.com mail adresine 11:45’ e kadar göndermeniz gerekmektedir.(Sonrasında gönderilen cevap

kağıtlarınızı kabul etmeyeceğim).Gönderdiğiniz mailin konusu şu şekilde olmalıdır: “Analiz4_Ad_Soyad” Örneğin,

Analiz4_Elif_Demir. Adınız, soyadınız, öğrenci numaranız ve imzanız cevap kağıdınızda bulunmalıdır (aksi takdirde

cevap kağıtlarınızı kabul etmeyeceğim).

Başarılar..